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Kendall L. Thomas

Education

Expected May 2027 **Ph.D., Statistics and Operations Research**
University of North Carolina at Chapel Hill, Chapel Hill, NC
Advisor: Dr. Jan Hannig
Dissertation: *Statistical Frontiers in Elite Athlete Performance*

Dec. 2025 **M.S., Statistics, Analytics, and Data Science**
University of North Carolina at Chapel Hill, Chapel Hill, NC

May 2019 **B.S., Mathematics; B.S., Computer Science**
Davidson College, Davidson, NC
magna cum laude

Research Interests

- Statistical inference for complex and structured data
- Sports analytics and data-driven decision-making
- Longitudinal wearable-sensor and movement data
- Generalized fiducial inference and uncertainty quantification
- Compositional and overdispersed count data

Professional Experience

Mar. 2025–Present **Statistical Consultant**
Boston Athletic Association, Remote
Advised by Dr. Richard Smith, University of North Carolina at Chapel Hill

Aug. 2019–Apr. 2022 **Software Engineer; promoted to Software Engineer II**
Microsoft, Azure Data, Redmond, WA

Aug. 2016–May 2019 **Research Assistant**
Dr. Tim Chartier, Davidson College, Davidson, NC

Jan. 2018–Jul. 2019 **Assistant Analyst**
John Tobias, ESPN and FOX Sports, Remote

May 2018–Aug. 2018 **Advanced Analytics and Machine Learning Intern**
Nike, Beaverton, OR

Feb. 2017–Nov. 2017 **Research Assistant**
Athlete Intelligence, Seattle, WA / Remote

Apr. 2015–Present **Math, Statistics, and Computer Science Tutor**
Wyzant, Remote

Research Experience

Fall 2023–Present **Graduate Research Assistant**
Dr. Jan Hannig, Principal Investigator, University of North Carolina at Chapel Hill
Chapel Hill, NC
Conduct research in generalized fiducial inference, uncertainty quantification, hybrid discrete–continuous count data, and statistical analysis of elite-athlete movement data. First author of a peer-reviewed publication, a submitted manuscript, and a manuscript in preparation, both under Dr. Hannig's supervision. Develop and implement statistical methodology, conduct simulation studies and real-data analyses, develop research software, and contribute to grant proposal development and preparation. Supported during Fall 2025 and Fall 2026 by NSF Grant Nos. DMS-2113404, DMS-2210337, and DMS-2515303, and by the United States–Israel Binational Science Foundation under Grant No. 2024055.

Mar. 2025–Present **Statistical Consultant**
Boston Athletic Association, Remote
Advised by Dr. Richard Smith, University of North Carolina at Chapel Hill
Analyze historical race-performance data and provide statistical advice on Boston Marathon qualifying-time cutoffs, time adjustments for substantial net-downhill courses, and related registration policies. Contributed to analyses informing implemented policies for the 2026 and 2027 Boston Marathons, including a net-downhill qualifying-time adjustment and an additional random-selection opportunity for qualifiers.

Aug. 2016–May 2019 **Research Assistant**
Dr. Tim Chartier, Davidson College, Davidson, NC
Conducted data-analytics research involving NBA and NCAA basketball and soccer. Developed analyses and visualizations of biometric and performance data for Davidson College's Women's Soccer team; collaborated with psychology faculty on statistical investigations of the "hot hand" phenomenon; and designed a basketball hustle statistic for NBA applications and potential adaptation to NCAA competition.

Jan. 2018–Jul. 2019 **Assistant Analyst**
John Tobias, ESPN and FOX Sports, Remote
Conducted statistical research and supported the development of in-broadcast statistics, visualizations, and analytical storylines for televised sports coverage, including the 2019 FIFA Women's World Cup.

Feb. 2017–Nov. 2017 **Research Assistant**
Athlete Intelligence, Seattle, WA / Remote
Integrated football and soccer game data with existing biometric datasets to support athlete-performance research. Identified performance metrics correlated with athletic outcomes, combined and visualized multimodal data to detect trends, and contributed to analyses of potential injury-risk indicators.

Peer-Reviewed Publications

- **Thomas, K. L.**, & Hannig, J. (2026). Movement dynamics in elite female soccer athletes: The Quantile Cube approach. *Journal of Quantitative Analysis in Sports*, 22(3), 245–262. https://doi.org/10.1515/jqas-2025-0014

Submitted Papers

- **Thomas, K. L.**, & Hannig, J. Generalized Fiducial Inference for Hybrid Discrete–Continuous Count Data: Theory and Application. Submitted to the *Annals of Applied Statistics*, October 2026. arXiv:2609.38611. Code: https://github.com/kelanethomas/hgfi-count-data.

Working Papers

- **Thomas, K. L.**, Duncombe, P., & Hannig, J. Precision in Motion: Extending the Quantile Cube Framework. Manuscript in preparation.

Presentations

*Presenting author

Invited Talks

Scheduled Nov. 2026 **"Precision in Motion: From Athlete to Analyst"**
Midwest Sports Analytics Meeting, Virtual
Keynote Speaker

Oct. 2026 **"Generalized Fiducial Inference for Hybrid Discrete–Continuous Count Data"**
International Conference on Advances in Interdisciplinary Statistics and Combinatorics (AISC), Lafayette, LA
Thomas, K. L.*, & Hannig, J.

Apr. 2026 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
Connecticut Sports Analytics Symposium, Mansfield, CT
Thomas, K. L.*, & Hannig, J.

Jan. 2026 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
American Soccer Insights Summit, Houston, TX
Thomas, K. L.*, & Hannig, J.

Apr. 2019 **"Beyond a Bit Fit"**
Carolina Sports Analytics Meeting, Greenville, SC
Thomas, K. L.*, & Chartier, T.

Conference Presentations

Aug. 2026 **"Does Age Shape How Athletes Move? Extending the Quantile Cube Framework to Detect Age Effects in Movement Distribution Profiles"**
Joint Statistical Meetings (JSM), Boston, MA
Topic-Contributed Session: "Statistical Methods for Estimating Player Performance Across Ages and Leagues."
Thomas, K. L.*, & Hannig, J.

Apr. 2026 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
ASA Virtual Sports Analytics Conference, Virtual
Thomas, K. L.*, & Hannig, J.

Nov. 2024 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
Midwest Sports Analytics Meeting, Virtual
Thomas, K. L.*, & Hannig, J.

Nov. 2019 **"Beyond a Bit Fit"**
Midwest Sports Analytics Meeting, Pella, IA
Thomas, K. L.*, & Chartier, T.

Conference Posters

Feb. 2026 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
CoreAI UNC Research Showcase, Chapel Hill, NC
Thomas, K. L.*, & Hannig, J.

Sep. 2025 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
New England Symposium on Statistics in Sports, Boston, MA
Thomas, K. L.*, & Hannig, J.

Nov. 2024 **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"**
Carnegie Mellon Sports Analytics Conference, Pittsburgh, PA
Thomas, K. L.*, & Hannig, J.

Jul. 2024 **"Professional vs. Elite Female Soccer Athletes: Physical and Technical Performance Characteristics"**
National Strength and Conditioning Association National Conference, Baltimore, MD
Cantú, E. I.*, Moore, S. R., Thomas, K. L., & Smith-Ryan, A. E.

Software

- `quantile-cube`: Python package for visualizing quantile-based movement distributions. Available at `https://pypi.org/project/quantile-cube/`.

Teaching

Curriculum Developed

"Introduction to Data Science in Python"

Led a curriculum redesign introducing a Python-based section of STOR 320: *Methods and Models of Data Science* (first offered Summer 2024; renamed from *Introduction to Data Science* in Fall 2025) alongside the existing R-based version. Developed new lectures and Jupyter-based lab modules emphasizing data acquisition, cleaning, visualization, modeling, and communication in Python. The Python section is now offered each semester, providing students an alternative track aligned with modern data science tools and reproducible workflows.

STOR 290: Sports Analytics Intelligence Lab — Co-developed with Dr. Mario Giacomazzo a new 3-credit course (first offered Fall 2025) by transforming the Sports Analytics Intelligence Lab (SAIL) into an official course offering. Students work on real projects for UNC athletic programs and external partners, requiring them to formulate research questions, select appropriate methods, interpret results, and communicate findings to non-technical audiences. Built the curriculum around staged research milestones with feedback and revision required at each stage before students progress to the next, alongside weekly data science lectures (Shiny, version control, model building, visualization, web scraping) and oral presentation requirements.

Courses Taught (Instructor of Record)

Fall 2026	STOR 290: *Sports Analytics Intelligence Lab* – 9 students
Summer Session II 2026	STOR 455: *Methods of Data Analysis* – 6 students
Spring 2026	STOR 320: *Methods and Models of Data Science* – 42 students
Summer Session II 2025	STOR 455: *Methods of Data Analysis* – 15 students
Spring 2025	STOR 320: *Introduction to Data Science* – 94 students
Fall 2024	STOR 455: *Methods of Data Analysis* – 101 students
Summer Session II 2024	STOR 320: *Introduction to Data Science* – 22 students

Lab Instructor

Fall 2023	STOR 120: *Introduction to Data Science*
Spring 2023	STOR 120: *Introduction to Data Science*

Teaching Assistant

Spring 2024	STOR 538: *Sports Analytics*
Fall 2022	STOR 455: *Methods of Data Analysis*

Fellowships, Grants, & Travel Awards

Oct. 2026 — **Member Initiative Grant, Principal Investigator**
American Statistical Association, Section on Statistics and Data Science Education
$400
Competitive award supporting the Sports Analytics Intelligence Lab (SAIL).

Jun. 2026 — **John McNeely DuBose Graduate Fellowship Fund**
University of North Carolina at Chapel Hill
$650

Jan. 2026 — **Raj Chandra Bose Graduate Student Travel Fund**
University of North Carolina at Chapel Hill
$1,000

Jan. 2026 — **The Graduate School Transportation Grant**
University of North Carolina at Chapel Hill
$400

Aug. 2025–May 2026 — **Druscilla French Graduate Student Excellence Fund Award**
University of North Carolina at Chapel Hill
$5,000

Dec. 2024 **Raj Chandra Bose Graduate Student Travel Fund**
University of North Carolina at Chapel Hill
$750

Awards and Honors

Academic

Spring 2026 **Boka W. Hadzija Award for Distinguished University Service Nominee**
Nominated by the Department of Statistics and Operations Research, University of North Carolina at Chapel Hill

Dec. 2024 **Walter Deemer Excellence in Teaching Award**
UNC Department of Statistics and Operations Research

Dec. 2023 **Excellence in Teaching Assistance and Instruction Award**
UNC Department of Statistics and Operations Research

Conference & Research Competitions

Sep. 2025 **First Place Poster Award**
New England Symposium on Statistics in Sports (NESSIS), Boston, MA

Nov. 2024 **First Place Poster Award**
Carnegie Mellon Sports Analytics Conference, Pittsburgh, PA

Mar. 2020 **Hackathon Finalist (Professional Division)**
MIT Sloan Sports Analytics Conference, Boston, MA

Oct. 2017 **Grace Hopper Celebration Research Scholar**
Computing Research Association – Women, Orlando, FL

Industry & Technical Hackathons

Fall 2021 **Winner – Greatest Business Value**
Microsoft Azure Supportability Hackathon

Spring 2021 **Winner – Most Impactful & Greatest Business Value**
Microsoft Azure Supportability Hackathon

Fall 2019 **Winner – Best Hack (Audience Vote)**
Microsoft Azure Data Hackathon

Leadership, Service, & Outreach

Aug. 2026–Present **Planning Committee Member**
American Soccer Insights (ASI) Summit

Jan. 2024–Present **Graduate Student Liaison**
UNC Department of Statistics and Operations Research

Aug. 2023–Present **Graduate Student Mentor**
UNC Department of Statistics and Operations Research

Jan. 2023–Present **Principal (appointed May 2025) and Graduate Student Advisor**
Sports Analytics Intelligence Lab (SAIL), UNC Department of Statistics and Operations Research
As Principal, lead the lab and oversee the organization and direction of its research agenda.

Jul. 2015–May 2019 **Varsity Athlete, NCAA Division I Women's Soccer Team**
Davidson College

Jan. 2018–May 2019 **Analyst & Team Manager, NCAA Division I Men's Basketball Team**
Davidson College

Jan. 2016–May 2019 **'Cats Stats Lead Analyst**
Davidson College

Aug. 2017–May 2019 **President, Females in Computer Science and Information Technology**
Davidson College

Professional Memberships

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- Association for Women in Mathematics (AWM)
- Phi Beta Kappa Society
- Omicron Delta Kappa: The National Leadership Honor Society

Technical Skills

Programming Languages

Advanced	Python, R
Intermediate	Java, HTML, CSS, SQL, Kusto
Basic	C++, C#, C, JavaScript, p5.js, d3.js

Software and Tools

Advanced	LaTeX, RStudio, Microsoft Office, Tableau, Jupyter Notebooks
Intermediate	SQL Server Management Studio, Azure Data Studio
Basic	Mathematica, ArcGIS, MATLAB

Libraries and Frameworks

Data Analysis	pandas, NumPy, SciPy
Visualization	ggplot2, matplotlib, Plotly, seaborn
Interactive Apps	Shiny (R and Python)
Machine Learning	scikit-learn, TensorFlow, PyTorch
Web Scraping	BeautifulSoup, requests, Selenium
Version Control	Git / GitHub